

Placenta-Aware Adaptive Support System for Modulating Uteroplacental Perfusion and Attenuating Gut–Placenta Inflammatory Signaling to Improve Fetal Growth

1. Introductory Information

Fetal Growth Restriction (FGR) affects over 23 million pregnancies annually and remains one of the most consequential yet mechanistically underaddressed complications of pregnancy. Defined by the failure of the fetus to reach its genetically determined growth potential — most often due to placental insufficiency — FGR carries elevated risks of perinatal mortality, intraventricular hemorrhage, and lifelong neurodevelopmental and metabolic vulnerability. Despite growing understanding of placental biology, clinical management remains largely passive: serial ultrasound surveillance without any targeting of the underlying pathophysiology. No scalable intervention currently exists that directly addresses the upstream biomechanical drivers of placental dysfunction.

Reduced uteroplacental perfusion initiates a cascade of hypoxia, oxidative stress, and cytokine-mediated inflammation that impairs trophoblast function and restricts fetal growth. Emerging evidence further implicates the maternal gut microbiome in amplifying this cycle — gut-derived lipopolysaccharides and microbial metabolites translocate into systemic circulation under inflammatory conditions, feeding back into placental immune activation through what is increasingly recognized as the gut–placenta axis. Mechanical compression of the gravid abdomen — from postural loading, gravity, or ill-fitting garments — represents an underappreciated, modifiable upstream driver of this entire cascade.

We propose a closed-loop adaptive wearable system that dynamically redistributes abdominal mechanical stress to restore uteroplacental perfusion and attenuate hypoxia-driven inflammatory signaling along the gut–placenta axis.

2. Proposal Information

Central Hypothesis

Sustained, position-dependent mechanical loading of the maternal abdomen compromises uteroplacental blood flow, potentiates hypoxic inflammatory signaling along the gut–placenta axis, and thereby restricts fetal growth. Conversely, adaptive redistribution of that mechanical load will restore perfusion, reduce cytokine-mediated dysfunction, and create biological conditions that support improved fetal outcomes.

Scientific Rationale and Supporting Evidence

The mechanistic logic driving this proposal is grounded in well-established physiology. Mechanical compression restricts uterine arterial flow, dropping oxygen delivery to the intervillous space. Trophoblasts respond to this hypoxia by upregulating HIF-1 α , which drives expression of pro-inflammatory cytokines — particularly IL-6 and TNF- α — that impair vascular remodeling and perpetuate perfusion deficits. Simultaneously, gut epithelial barrier integrity is compromised under these inflammatory conditions, permitting microbial translocation that amplifies placental immune activation. The gut–placenta axis is therefore not a parallel pathway but an amplifier of the primary biomechanical insult. The conceptual parallel to pressure redistribution in advanced wound care — a well-validated clinical intervention — supports the biological plausibility of the proposed approach.

Proposed Work and Methodology

Work is organized across four sequential phases spanning 18 months, each gated by pre-specified go/no-go criteria before resource investment in the next phase.

Phase 1 — Device Engineering (Months 1–3): Flexible pressure-mapping sensor arrays, dynamically inflatable support cells distributed across the abdominal surface, and a microcontroller-driven feedback loop will be integrated into a functional prototype. The system continuously adjusts support geometry in response to real-time pressure data. Mechanical performance will be validated on abdominal phantoms to confirm spatial pressure resolution before any biological work begins.

Phase 2 — In Vitro Mechanistic Validation (Months 4–8): Placental explants will be subjected to controlled compression in bioreactor chambers. Primary trophoblast and endothelial cell cultures will be exposed to hypoxia

gradients mimicking compression-induced ischemia, with cytokine secretion profiles (IL-6, TNF- α) and oxidative stress markers (ROS, NRF2, 8-OHdG) measured under simulated device-on and device-off conditions. Gut epithelial permeability assays — including transepithelial electrical resistance and LPS translocation — will directly interrogate the gut–placenta signaling axis.

Phase 3 — Preclinical In Vivo Studies (Months 9–14): A validated ovine or rodent FGR model will be used to evaluate device effects on uteroplacental Doppler flow indices, placental histology, and fetal biometric outcomes. Tissue-level cytokine profiling and oxidative stress assays at study endpoints will confirm in vivo mechanistic alignment with in vitro findings. Preclinical success criteria require $\geq 20\%$ improvement in uteroplacental blood flow velocity by Doppler ultrasound and $\geq 25\%$ reduction in placental IL-6 and TNF- α concentrations relative to untreated controls.

Phase 4 — Human Feasibility (Months 15–18): A prospective observational cohort will enroll 40–50 pregnant women at elevated risk for FGR. Safety, adherence, comfort, and preliminary changes in Doppler parameters and serum biomarkers (IL-6, TNF- α , LPS, zonulin) will be assessed. This phase is not powered for efficacy; its purpose is to generate the tolerability and early-signal data required to justify a full randomized controlled trial. Feasibility success is defined as $\geq 80\%$ completion without serious device-attributable adverse events.

3. Path to Impact

The development pathway follows a structured progression from prototype to validation to human deployment, with each stage designed to generate the evidence base required for the next. Phase 1 delivers a functional prototype validated under mechanical testing conditions. Phase 2 produces mechanistic in vitro evidence linking abdominal pressure redistribution to reduced placental inflammatory signaling — providing the scientific foundation for animal work. Phase 3 translates those in vitro findings into a living system, generating the Doppler and histological data necessary to justify human exposure. Phase 4 establishes the feasibility dataset — safety profile, adherence rates, device comfort, and early biomarker signals — required to power and justify a full-scale randomized controlled trial.

Model systems are selected to match the biological question at each phase: in vitro trophoblast and gut epithelial cultures for mechanistic signal detection; ovine or rodent FGR models for in vivo confirmation (chosen for established translational relevance to human placental physiology); and a 40–50 participant human feasibility cohort drawn from high-risk antenatal populations. Sample size for the feasibility cohort is sufficient to detect clinically relevant signals in Doppler parameters and serum biomarkers while maintaining ethical proportionality prior to full trial justification.

The longer-term pathway beyond this grant leads to a multi-site randomized controlled trial powered for fetal growth and neonatal outcomes, followed by regulatory pathway development and scaled manufacturing. The device concept has been intentionally designed with global scalability as a primary engineering constraint. It requires no continuous clinical supervision, no pharmacological supply chains, and no advanced imaging infrastructure. Core components leverage low-cost flexible electronics and textile fabrication techniques amenable to local manufacturing — avoiding the supply-chain dependencies that have historically limited technology adoption in resource-limited settings.

FGR disproportionately affects pregnancies in low- and middle-income countries — particularly across South Asia and sub-Saharan Africa — where it accounts for the majority of preventable perinatal deaths and where existing management strategies are functionally unavailable. This system integrates naturally into existing antenatal care workflows and can be supported by trained community health workers, making it practically deployable in the settings where the burden of FGR is most acute and the absence of any effective intervention most consequential.

4. Team Capabilities

The team brings together complementary expertise across biomedical engineering, maternal-fetal medicine, placental biology, and global health implementation — a combination directly matched to the scientific and translational demands of this project. Engineering capacity covers wearable systems design, flexible sensor integration, embedded firmware development, and mechanical prototype validation. Clinical expertise includes high-risk obstetrics, antenatal surveillance, and experience managing FGR in both hospital and community settings. Research capacity spans cell culture and

bioreactor systems, in vivo rodent and ovine models, cytokine and oxidative stress assay platforms, and Doppler ultrasound methodology.

Critically, the team has direct experience working in resource-limited settings across South Asia and sub-Saharan Africa, including prior involvement in antenatal care program design, community health worker training, and technology deployment under infrastructure constraints. This experience has directly shaped the engineering choices made here — the avoidance of cold-chain dependencies, the emphasis on locally manufacturable components, and the integration with existing care workflows rather than parallel infrastructure. The team is positioned not only to conduct the science but to ensure that what it produces is implementable in the environments where it is most needed.